

A. Train/test split and mechanism-label distribution

Dataset split	Genes included	Total variants	LoF	GoF	LoF (%)	GoF (%)
Training set (All non-SCN1A)	SCN1A sodium-channel genes	287	161	126	56.1	43.9
Held-out test set (SCN1A only)	SCN1A only	61	49	12	80.3	19.7
Full validated dataset	All sodium-channel genes	348	210	138	60.3	39.7

B. Gene-level breakdown of validated GoF/LoF variants

Gene	Split	Total	LoF	GoF	LoF (%)	GoF (%)
SCN1A	Held-out test	61	49	12	80.3	19.7
SCN5A	Training	157	117	40	74.5	25.5
SCN2A	Training	32	14	18	43.8	56.2
SCN4A	Training	31	13	18	41.9	58.1
SCN8A	Training	22	8	14	36.4	63.6
SCN10A	Training	16	7	9	43.8	56.2
SCN3A	Training	12	2	10	16.7	83.3
SCN9A	Training	11	0	11	0.0	100.0
SCN11A	Training	6	0	6	0.0	100.0

LoF = loss of function; GoF = gain of function. SCN1A was held out as the main test gene.